

OSSP (Operating Systems And Systems Programming) - 25CS2104E

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

(uname):

```
advaith@DESKTOP-0E10BMR:~$ nano childprocess.c
advaith@DESKTOP-0E10BMR:~$ gcc childprocess.c -o program
advaith@DESKTOP-0E10BMR:~$ ./program
Enter Linux command: uname
Parent PID : 1152
Child PID : 1153
Linux
Child process completed.
```

(lscpu):

```

Child process completed.
advaith@DESKTOP-0E10BMR:~$ gcc childprocess.c -o program
advaith@DESKTOP-0E10BMR:~$ ./program
Enter Linux command: lscpu
Parent PID : 1183
Child PID : 1184
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      8
On-line CPU(s) list:        0-7
Vendor ID:                   GenuineIntel
Model name:                  Intel(R) Core(TM) i7-1065G7 CPU @ 1.30GHz
CPU family:                  6
Model:                      126
Thread(s) per core:         2
Core(s) per socket:         4
Socket(s):                   1
Stepping:                    5
BogoMIPS:                    2995.20
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge m
ca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht sysc
all nx pdpe1gb rdtscp lm constant_tsc arch_perfmon rep
_good nopl xtopology tsc_reliable nonstop_tsc cpuid ts
c_known_freq pni pclmulqdq vmx ssse3 fma cx16 pdcm pci
d sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer
aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dno
wprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shado
w ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep b
mi2 erms invpcid avx512f avx512dq rdseed adx smap avx5
12ifma clflushopt avx512cd sha_ni avx512bw avx512vl xs
aveopt xsavec xgetbv1 xsaves vnmi avx512vbmi umip avx5
12_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bital
g avx512_vpopcntdq rdpid fsrm md_clear flush_l1d arch_
capabilities

Virtualization features:
Virtualization:              VT-x
Hypervisor vendor:           Microsoft
Virtualization type:         full
Caches (sum of all):
L1d:                         192 KiB (4 instances)
L1i:                         128 KiB (4 instances)
L2:                           2 MiB (4 instances)
L3:                           8 MiB (1 instance)
NUMA:
NUMA node(s):                1
NUMA node0 CPU(s):          0-7
vulnerabilities:
Gather data sampling:        Unknown: Dependent on hypervisor status
Ghostwrite:                  Not affected
Indirect target selection:   Mitigation; Aligned branch/return thunks
Itlb multihit:               KVM: Mitigation: Split huge pages
L1tf:                        Not affected
Mds:                         Not affected
Meltdown:                    Not affected
Mmio stale data:             Vulnerable: Clear CPU buffers attempted, no microcode;
                              SMT Host state unknown
Old microcode:               Not affected
Reg file data sampling:      Not affected
Retbleed:                    Mitigation; Enhanced IBRS
Spec rstack overflow:        Not affected
Spec store bypass:           Mitigation; Speculative Store Bypass disabled via prct
                              l
Spectre v1:                  Mitigation; usercopy/swapgs barriers and __user pointe
                              r sanitization
Spectre v2:                  Mitigation; Enhanced / Automatic IBRS; IBPB conditiona
                              l; PBRBSB-eIBRS SW sequence; BHI SW loop, KVM SW loop
Srbds:                       Unknown: Dependent on hypervisor status
Tsa:                         Not affected
Tsx async abort:             Not affected
Vmscape:                     Not affected
Child process completed.
advaith@DESKTOP-0E10BMR:~$ -|

```

(lsblk):

```
advaith@DESKTOP-0E10BMR:~$ gcc childprocess.c -o p
advaith@DESKTOP-0E10BMR:~$ ./program
Enter Linux command: lsblk
Parent PID : 1238
Child PID : 1239
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0     2G   0 disk [SWAP]
sdd   8:48   0     1T   0 disk /mnt/wslg/distro
/

Child process completed.
advaith@DESKTOP-0E10BMR:~$ |
```

(ps):

```
advaith@DESKTOP-0E10BMR:~$ gcc childproce
advaith@DESKTOP-0E10BMR:~$ ./program
Enter Linux command: ps
Parent PID : 1265
Child PID : 1266
  PID TTY          TIME CMD
  314 pts/0        00:00:00 bash
   804 pts/0        00:00:00 nano
  1265 pts/0        00:00:00 program
  1266 pts/0        00:00:00 ps
Child process completed.
advaith@DESKTOP-0E10BMR:~$ |
```

(top):

```
top - 09:43:53 up 53 min,  1 user,  load average: 0.07, 0.02, 0.00
Tasks: 26 total,  1 running, 24 sleeping,  1 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 98.1 id,  1.9 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  7819.6 total,  7207.5 free,    518.0 used,   244.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,    0.0 used. 7301.6 avail Mem

   PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24256   15344  11524 S   0.0   0.2   0:00.77 systemd
    2 root        20   0    3180    2208   2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0    3180    2044   1940 S   0.0   0.0   0:00.00 init
   47 root       19  -1   50392   16876  15592 S   0.0   0.2   0:00.26 systemd-journal
   79 systemd+    20   0   22540   14636  12016 S   0.0   0.2   0:00.09 systemd-resolve
   88 root       20   0   35324   12352   9300 S   0.0   0.2   0:00.58 systemd-udev
  108 root       20   0    2888    2012   1884 S   0.0   0.0   0:00.04 chronyd-starter
  109 root       20   0    4472    3212   2964 S   0.0   0.0   0:00.01 cron
  110 message+   20   0    8800    5472   4796 S   0.0   0.1   0:00.07 dbus-daemon
  114 root       20   0   44092   29844  14704 S   0.0   0.4   0:00.26 networkd-dispat
  136 root       20   0   18432    9464   8220 S   0.0   0.1   0:00.10 systemd-logind
  168 syslog     20   0  220548    5964   4668 S   0.0   0.1   0:00.08 rsyslogd
  185 root       20   0    5492    2832   2644 S   0.0   0.0   0:00.00 agetty
  194 root       20   0  123456   32676  18064 S   0.0   0.4   0:00.24 unattended-upgr
  199 _chrony     20   0   24456  11968   7188 S   0.0   0.1   0:00.14 chronyd
  209 _chrony     20   0   12092    2464   1824 S   0.0   0.0   0:00.01 chronyd
  312 root       20   0    3184    1112    980 S   0.0   0.0   0:00.00 SessionLeader
  313 root       20   0    3200    1252   1108 S   0.0   0.0   0:00.47 Relay(314)
  314 advaith    20   0    6784    6128   3928 S   0.0   0.1   0:00.61 bash
  315 root       20   0    8648    5640   4796 S   0.0   0.1   0:00.01 login
  358 advaith   20   0   22336   13620  10992 S   0.0   0.2   0:00.13 systemd
  360 advaith   20   0   22908    4128   2128 S   0.0   0.1   0:00.00 (sd-pam)
  388 advaith   20   0    6136    5456   3916 S   0.0   0.1   0:00.03 bash
   804 advaith   20   0    4328    3324   2624 T   0.0   0.0   0:00.00 nano
  1290 advaith   20   0    2768    1944   1848 S   0.0   0.0   0:00.00 program
  1291 advaith   20   0    8180    5732   3636 R   0.0   0.1   0:00.00 top
```